



Universität
Zürich^{UZH}



Blockchain & DLT
Research Group



Network Science HS26

Assignment

Blockchain & Distributed Ledger Technologies Group

FOR STUDIES PURPOSES ONLY

UZH Blockchain Center
Faculty of Business, Economics and Informatics
University of Zurich
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Bioinformatics Network Investigation

You are given an undirected network representing an enzyme structural graph.

Your task is to investigate the structure of this network and develop an evidence-based description of its organization. Your analysis should use concepts and methods discussed in the course.

You are **not expected to follow a prescribed sequence of analyses**. Decide which measurements, comparisons, and visualizations are useful, and explain why they help you understand the network.

Network Data Format

The supplied network is provided as an **edge list**.

Each row of the file represents one edge and contains two node identifiers:

node_1	node_2
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For example,

1	7
1	12
2	5

means that nodes 1 and 7 are connected, nodes 1 and 12 are connected, and nodes 2 and 5 are connected.

The network is **undirected**, so an edge between nodes i and j is the same as an edge between nodes j and i .

The two directions are therefore not listed separately.

Each node identifier refers to a node in this particular network. The identifiers themselves do not imply that one node is more important than another.

You should load the edge list and construct the network before beginning your analysis.

Investigation

Your task is to investigate the network and develop an evidence-based description of its structure. The following questions indicate aspects that may be useful to consider. They are intentionally open-ended, and you are not expected to address them in the order given.

- **Connectivity and structure:** What does the network look like at the global level? Are there features of its connectivity or topology that are important for understanding the network?
- **Important or unusual nodes:** Are some nodes structurally more important or unusual than others? Choose appropriate measures of centrality or ranking, identify interesting nodes, and explain what makes them important or unusual. Consider whether different notions of importance lead to the same or different nodes.
- **Degree structure:** What does the degree distribution reveal that cannot be inferred from the average degree alone?
- **Relationships between connected nodes:** Do nodes tend to connect to other nodes with similar or different degrees? What does this tell you about the network's organization?
- **Local structure:** What do triangles and clustering reveal about how the network is organized locally? Use examples from the actual network to support your interpretation.

- **Communities and mesoscopic structure:** Does the network contain groups of nodes that are more strongly connected to one another than to the rest of the network? If so, how convincing is this structure, and which nodes or connections are particularly relevant to your interpretation?
- **Network characteristics:** Based on your observations, what characteristics associated with random, small-world, or scale-free networks, if any, are evident in this network? Discuss the evidence for and against these characteristics, and explain the limitations of your conclusions.
- **Beyond the summary:** Identify at least **one structural feature that would be difficult to discover from the basic network statistics alone**. Show how you discovered it and explain why it matters.

Your analysis should not simply report numerical values. For important results, explain **what the result means for this particular network** and support your interpretation with evidence from the network.

Deliverables

Submit:

1. **Written report** describing your investigation and conclusions.
2. **Code/notebook** used to perform the analysis.
3. **Supporting visualizations and calculations** required to reproduce and verify your conclusions.

Your code should be sufficiently clear that the analysis can be reproduced from the supplied network.

Important Requirements

- Use the supplied network as the basis for your analysis.
- Do not rely on precomputed network statistics as given answers. Obtain and verify relevant quantities from the supplied network yourself.
- Distinguish between **observation, measurement, and interpretation**.
- When identifying an interesting node or structural feature, give enough information for it to be located in the original network.
- Do not rely on a single network measure to characterize the network.
- Where you make comparisons with another network or network model, explain how the comparison was chosen, which properties were controlled for, what was compared, and why the comparison is appropriate.
- Your conclusions must be supported by evidence from the actual network rather than by general descriptions of network-science concepts.
- Your report should contain at least **three claims that depend on inspecting and analyzing the supplied network itself**, rather than simply on knowing the definitions of network measures.
- At least one of these claims must refer to specific **node(s), edge(s), or group(s)** in the supplied network.

IMPORTANT: The goal is not to calculate as many network measures as possible. A strong analysis selects appropriate methods, uses evidence from the network, and explains what the results reveal about its structure.

Reproducibility

Your submitted notebook or code should contain all steps necessary to reproduce the main results of your report, starting from the supplied edge-list file.

All reported numerical results and visualizations should be traceable to your code. The final notebook should be executed so that the relevant outputs are visible.